

MTH2021 Mathematics Methods 2

Self-study sheet 1

1. If $z_1 = 2 + 3i$, $z_2 = 1 - 2i$, evaluate $z_1 + 2z_2$, $z_1 z_2$ and z_1/z_2 . Find the modulus and the principal value of the argument of z_1/z_2 .
2. (a) Let $\arg z = 1237.2\pi$. Find $\text{Arg } z$.
(b) Let $z = \sqrt{3} + i$. Find the modulus and the principal value of the argument of z^8 .

3. (a) Solve the equations

$$z^3 = -27i \quad z^3 = 1 \quad z^4 = i$$

for z .

- (b) Find all solutions of the equation

$$z(1 + z + z^2 + z^3 + z^4) = 0$$

4. Sketch the curves/regions represented by the following equations/inequalities:

- (a) $z = t + it^3$ for $-\infty < t < +\infty$

- (b) $|z - i|^2 + |z + i|^2 = 4$

- (c) $|z - i| \leq 2$

5. Let $z = x + iy \neq 0$. Define z^{-1} by $zz^{-1} = 1$.

- (a) By writing $z^{-1} = a + ib$, find z^{-1} and show that it is unique.

- (b) Show that $(z^{-1})^* = (z^*)^{-1}$.

- (c) Show that $(z_1/z_2)^* = z_1^*/z_2^*$. (Assume $(z_1 z_2)^* = z_1^* z_2^*$ – this relation in turn is a homework exercise.)

6. Derive the relations

- (a)

$$\exp(i\theta) + \exp(-i\theta) = 2 \cos \theta$$

$$\exp(i\theta) - \exp(-i\theta) = 2i \sin \theta$$

- (b)

$$\frac{\exp(i\theta) - 1}{\exp(i\theta) + 1} = i \tan(\theta/2)$$

- (c)

$$\frac{1}{\exp(i\theta) - 1} = -\frac{1}{2} + \frac{1}{2i} \cot(\theta/2)$$

7. Use De Moivre's theorem to evaluate $(1 + i)^{20}$.
-